

Market Basket Analysis and Association Rule Mining

Evidence from a Large-Scale Online Grocery Dataset

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Abstract

This report investigates customer co-purchasing behavior in an online grocery platform using Market Basket Analysis and Association Rule Mining (ARM). A computationally feasible subset of the Instacart dataset was constructed by sampling 20,000 users and focusing on historical (prior) orders. After cleaning and transaction construction, the Apriori algorithm was applied under two minimum support thresholds to identify frequent itemsets, followed by association rule extraction and ranking by lift. The resulting rules reveal statistically meaningful product affinities that can be operationalized in recommendation modules, promotional bundling, digital shelf design, and targeted marketing, with lift providing a robust indicator of non-random associations beyond baseline product popularity.

1 Introduction

Electronic commerce platforms increasingly rely on data-driven methods to understand purchasing behavior and improve customer experience. Market Basket Analysis (MBA) captures co-occurrence relationships between products purchased within the same transaction. Association Rule Mining (ARM) formalizes these relationships as rules of the form $X \rightarrow Y$, which can support cross-selling, bundling, and personalization strategies in online retail.

This report applies ARM to a large-scale online grocery dataset (Instacart) to extract interpretable purchasing rules and provide actionable business insights.

2 Data Scope and Transaction Definition

The Instacart dataset contains order histories and product identifiers at scale. To balance computational feasibility with statistical reliability, a subset of users was selected.

2.1 Subset selection

A random sample of **20,000 users** was drawn. Only *prior* orders were used to represent historical purchasing behavior.

2.2 Transaction definition

Each order corresponds to a single transaction (shopping basket):

- Transaction ID: `order_id`
- Items: `product_id` values appearing within the same order

After preprocessing, the analysis dataset contained:

- **300,374** valid transactions (orders with at least two distinct items)
- **3,176,865** product–order records
- Average basket size: **10.5763648** items

3 Data Preprocessing

Reliable association mining requires clean transaction logs. The following steps were applied:

1. **Missing value handling:** Rows with missing transaction or product identifiers were removed.
2. **Duplicate removal:** Duplicate (`order_id`, `product_id`) pairs were dropped to avoid inflating basket contents.
3. **Single-item transactions removed:** Orders containing only one product were excluded, as ARM requires co-occurrence.
4. **Returned items and price constraints:** The dataset does not provide explicit return indicators (e.g., negative quantity) nor product prices; therefore, return-based or price-based filtering is not applicable and is explicitly acknowledged.
5. **Rare item filtering:** To reduce dimensionality and improve Apriori tractability, items with frequency below 1% of transactions were removed. The resulting one-hot encoded matrix contained **108** frequent products.

4 Basket Construction

Transactions were transformed into a one-hot encoded matrix where rows represent orders and columns represent products. Values indicate item presence (binary).

Table 1: Transaction Matrix Dimensions

Description	Rows (Transactions)	Columns (Products)
Before rare-item filtering	300,374	41,189
After rare-item filtering (support ≥ 0.01)	300,374	108

5 Frequent Itemset Mining (Apriori)

The Apriori algorithm was executed under two minimum support thresholds to evaluate sensitivity:

Table 2: Frequent Itemsets under Different Minimum Support Thresholds

Min Support	Number of Itemsets	Max Itemset Size
0.01	125	2
0.05	6	1

5.1 Support selection rationale

At `min_support = 0.05`, only single-item sets were identified, which prevents association rule generation (no item pairs). Therefore, `min_support = 0.01` was selected as the final threshold, enabling discovery of co-purchased item pairs while remaining computationally feasible.

5.2 Why larger `min_support` yields fewer itemsets

Support thresholds act as a frequency filter: increasing `min_support` eliminates itemsets that do not occur sufficiently often. Due to Apriori's downward-closure property, when an itemset is infrequent, all of its supersets are pruned. Consequently, higher thresholds yield substantially fewer frequent itemsets, and the maximum itemset length typically decreases.

6 Frequent Items Visualization

Figure 1 presents the most frequently purchased items in the analysis subset. This plot provides context for interpreting rules: very popular items may appear in many baskets, inflating confidence values, which makes lift essential for evaluating non-random associations.

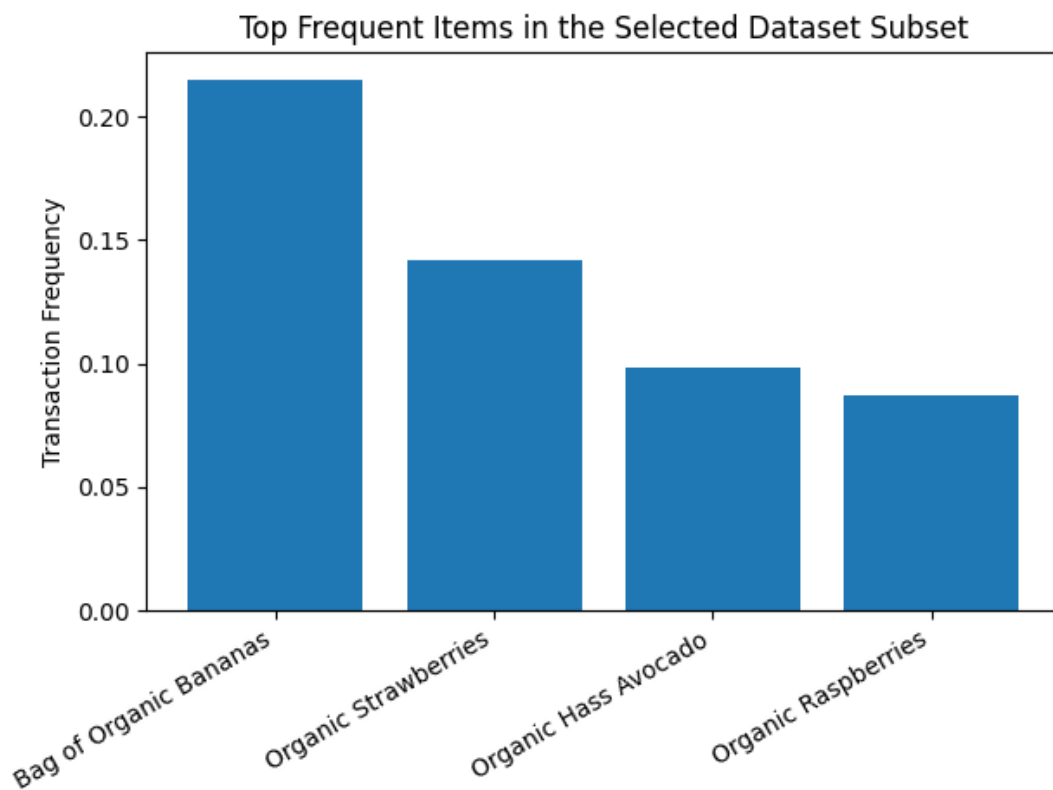


Figure 1: Top frequent items (by transaction frequency) in the selected dataset subset.

7 Association Rule Extraction

Association rules were generated from frequent itemsets using three key metrics:

- **Support:** proportion of transactions containing $X \cup Y$

- **Confidence:** conditional probability $P(Y | X)$
- **Lift:** $\frac{P(Y|X)}{P(Y)}$, indicating departure from independence

A total of **32** rules were produced. Table 3 reports the top rules ranked by lift.

7.1 Top rules by lift

Table 3: Top 3 Association Rules Ranked by Lift

Antecedent	Consequent	Support	Confidence	Lift
Organic Strawberries (21137)	Organic Raspberries (27966)	0.011 206	0.128 444	2.765 479
Bag of Organic Bananas (13176)	Organic Hass Avocado (47209)	0.021 240	0.171 110	2.394 118
Bag of Organic Bananas (13176)	Organic Raspberries (27966)	0.013 653	0.109 988	2.368 105

7.2 Interpretation of high-lift rules

The strongest association observed is between *Organic Strawberries* and *Organic Raspberries*, exhibiting the highest lift value (2.77). This indicates that customers purchasing organic strawberries are nearly three times more likely to also purchase organic raspberries compared to the baseline probability of purchasing raspberries. Additionally, *Bag of Organic Bananas* shows strong associations with both *Organic Hass Avocado* and *Organic Raspberries*, suggesting a broader organic-produce co-purchasing pattern. These rules have moderate support, indicating applicability at scale while capturing non-random behavior.

8 Business Interpretation

The extracted associations provide actionable signals for online grocery operations.

8.1 Product recommendations

Rules with lift substantially greater than 1 indicate that the consequent becomes more likely when the antecedent is present. These rules can power:

- “Frequently bought together” modules on product pages
- cart-level cross-sell suggestions (e.g., “Add Organic Raspberries?”)
- checkout-stage upsell prompts

8.2 Bundling and promotions

High-lift pairs are suitable candidates for:

- bundle discounts (e.g., strawberries + raspberries)
- conditional promotions (“Buy bananas, get avocados % off”)

Support should be monitored to ensure promotions target sufficiently common behavior rather than niche patterns.

8.3 Merchandising and digital shelf design

Associations can inform product adjacency in:

- category browsing flows
- search result reranking
- related product panels

8.4 Personalized marketing

If a customer frequently purchases *Bag of Organic Bananas*, targeted offers for *Organic Hass Avocado* or *Organic Raspberries* can improve conversion through personalized email or push notifications.

8.5 Inventory and demand planning

Complementary demand can be anticipated: if *Bag of Organic Bananas* is promoted or faces shortage, demand for associated products such as *Organic Hass Avocado* may shift accordingly. Rules can therefore support coordinated inventory decisions.

8.6 Reliability and risk

Lift captures association strength beyond popularity, but support indicates scale. Very high lift with very low support may reflect narrow segments; operational deployments should balance lift with minimum support constraints and consider A/B testing for validation.

9 Discussion

9.1 Role of lift

Lift is a central metric because it distinguishes genuine associations from popularity effects. Confidence may appear high when the consequent is generally popular; lift adjusts for the baseline frequency of the consequent:

- Lift = 1: independence (no meaningful association)
- Lift > 1: positive association (useful for cross-sell and bundling)
- Lift < 1: negative association (potential substitution or avoidance)

9.2 Why Apriori remains important for online retail

Apriori is valuable because it provides interpretable, auditable rules that are easy to communicate to business stakeholders. It supports core e-commerce functions such as recommendation, bundling, and merchandising. Even when more complex models exist, Apriori offers a transparent baseline and a practical starting point for market basket insights.

10 Conclusion

Using a large but tractable subset of an online grocery dataset, this report demonstrated an end-to-end ARM pipeline: preprocessing, basket construction, frequent itemset mining with Apriori, and rule extraction. The results revealed meaningful co-purchase affinities (high lift), which can be operationalized in recommendation systems, promotion design, and digital merchandising. Future extensions may incorporate temporal segmentation, aisle-level aggregation, and evaluation of uplift in controlled experiments.

A Appendix: Reproducibility Notes

- The analysis uses prior orders from a random sample of 20,000 users.
- Single-item transactions were excluded to ensure valid co-occurrence mining.
- Rare items were filtered using a 1% frequency threshold to ensure computational feasibility.
- The frequent-items chart should be exported from the notebook as `top_frequent_items.png` and placed in the same directory as this `.tex` file.